Thesis Progress Summary and Proposal: Signal Informativeness Sweep

1. Purpose of This Document

This report serves two purposes: (i) to summarise the experimental findings of the thesis as of April 2026, and (ii) to propose a follow-up experiment — a Signal Informativeness Sweep — designed to test a remaining boundary condition of the thesis's central claim. I would welcome your feedback on both the framing of the completed work and the necessity of the proposed experiment before committing compute resources.

2. Central Thesis Claim

The thesis investigates whether a regime-aware PPO market-making agent statistically outperforms a regime-blind PPO agent in a synthetic high-frequency environment. The central finding is a well-defended null result, framed by the following claim:

> *Explicit regime labels provide no marginal benefit when the observation space already contains an informative continuous volatility proxy (sigma_hat).*

This claim has been defended through five layers of evidence, summarised below.

3. Completed Experiments and Findings

#	Experiment	Test	Key Result	Conclusion
1	Main OOS (aware vs. blind, 20 seeds)	Paired t-test (Sharpe)	$p = 0.261$	No significant difference
1	Main OOS (aware vs. blind, 20 seeds)	Paired t-test (Equity)	$p = 0.023$	Minor equity advantage for blind
2	5-variant pure ablation (sigma_only, regime_only, combined, oracle_pure, oracle_full; 20 seeds)	Paired t-test (Sharpe)	sigma_only vs. oracle_full: $p = 0.115$	Oracle paradox: perfect labels do not beat sigma_only
3	Detector robustness (rv_baseline 60.7%, rv_dwell 60.4%, HMM 81.8%; 3 × 20 seeds = 120 models)	One-way ANOVA across detectors	$F \approx 0.003$, $p = 0.997$	Null result holds across detector quality
4	Regime-conditional inventory penalty ($\eta_H = 5 \cdot \eta_L$; 20 seeds)	Paired t-test (Sharpe)	sigma_only vs. combined: $p = 0.0016$	Regime-dependent reward does not rescue explicit labels

#	Experiment	Test	Key Result	Conclusion
5	Model misspecification (fill parameters A, k regime-linked; 20 seeds)	Paired t-test (Sharpe)	sigma_only vs. oracle_full: p = 0.881	Null result robust to environmental misspecification
6	TOST equivalence analysis (17 April 2026)	Two one-sided tests	Normal env: p = 0.0005 (±0.10 bound); Misspec env: p = 0.039 (±0.05 bound)	Active evidence of practical equivalence, not merely absence of difference

Secondary finding: Both PPO variants outperform the Avellaneda-Stoikov and naive fixed-spread baselines on risk-adjusted performance by a factor of 4–6× in Sharpe. The RL framework itself adds clear value; only the explicit regime label does not.

4. Why the Null Result Is Defensible

The null result rests on five independent layers of evidence:

1. **Mechanistic explanation.** sigma_hat, as the second element of the 6-dimensional observation vector, implicitly encodes regime information; the one-hot label is therefore redundant.
2. **Upper-bound test.** Even oracle agents with ground-truth regime access do not surpass sigma_only (oracle paradox).
3. **Detector quality invariance.** The result holds across three detectors spanning 60.4% to 81.8% accuracy.
4. **Reward-channel invariance.** Making the reward regime-sensitive does not make the label useful.
5. **Environmental robustness.** The result survives mild model misspecification in fill dynamics.

The addition of the TOST analysis converts this from a passive "no evidence of difference" into active evidence of practical equivalence within a ±0.10 Sharpe bound — a threshold corresponding to approximately 13% relative deviation around the observed Sharpe baseline of ~0.75.

5. The Remaining Open Question

Despite the breadth of the robustness checks above, one boundary condition has not yet been tested: **what happens when sigma_hat itself is no longer informative?**

The signal redundancy argument makes a falsifiable prediction: if explicit labels are redundant because sigma_hat already carries the regime signal, then progressively degrading sigma_hat should progressively increase the marginal value of the explicit label. This prediction has not yet been tested, and without it the claim remains categorical ("labels are redundant") rather than conditional ("labels are redundant when the continuous proxy is informative enough").

6. Proposed Experiment: Signal Informativeness Sweep

6.1 Research Question

Does the marginal value of the explicit regime label increase as sigma_hat becomes less informative?

6.2 Design

Five conditions, each progressively degrading sigma_hat. In each condition, the full five-variant ablation (sigma_only, regime_only, combined, oracle_pure, oracle_full) is re-trained with 20 seeds.

Condition	Manipulation	Motivation
clean	sigma_hat unchanged	Baseline (matches existing ablation)
noisy	Gaussian noise added to sigma_hat	Simulates estimation error
lagged	sigma_hat delayed by k steps	Simulates observation delay
coarsened	sigma_hat discretised to 3 levels	Simulates low-resolution data
none	sigma_hat removed entirely	Limiting case; forces reliance on alternative channels

Total: 5 conditions × 5 variants × 20 seeds = **500 models.**

6.3 Falsifiable Prediction

Under the signal redundancy hypothesis, the Sharpe gap between sigma_only and oracle_full should widen monotonically from clean to none. In the none condition, the explicit regime label should become essential rather than redundant, and oracle_full should significantly outperform variants lacking it.

6.4 What the Experiment Would Resolve

- **If the prediction is confirmed:** The signal redundancy argument is converted from a categorical claim into a conditional one, with an empirically mapped boundary. The null result is no longer merely defended — it is explained and localised.
- **If the prediction is disconfirmed:** The signal redundancy interpretation requires revision. The agent must be extracting regime-relevant information from a third channel (e.g. inventory dynamics, fill rates), which would itself be a novel and publishable finding.
- **If the result is partial:** We gain a practical map of which types of signal degradation (noise, lag, coarsening) make explicit labels valuable — a result with direct relevance to practitioners.

In all three cases, the experiment produces a useful outcome. This is a feature of the design.

6.5 Compute and Timeline

Based on prior runs (misspecification experiment: ~50 minutes per model on CPU), the full sweep is estimated at approximately 15–16 days of CPU time. I therefore propose a staged approach:

1. **Pilot phase (~2.3 days):** 3 seeds × 5 conditions × 5 variants = 75 models. The pilot is sufficient to detect the predicted monotonic trend if it exists.
2. **Decision point:** Based on pilot results, either commit to the full 20-seed run or reassess the design.
3. **Potential GPU acceleration:** An RTX 4070 SUPER is available locally; a short GPU benchmark during the pilot would determine whether GPU execution can reduce the full-run time meaningfully.

6.6 Scope Clarification

This sweep is proposed as the final robustness check of the thesis's central argument. I am not proposing to extend the thesis beyond it. Deeper follow-ups — information-theoretic formalisation of redundancy (mutual information estimates), real-data validation, and a strong misspecification variant — are intended for a post-thesis paper and would be discussed with you separately in that context.

7. Request for Feedback

I would particularly welcome your view on three questions:

1. Is the five-condition design appropriate, or would a narrower set (e.g. only clean and none as limiting cases) be preferable for the thesis timeline?
2. Is the ±0.10 equivalence bound in the TOST analysis defensible as a primary threshold, or should I report multiple bounds as sensitivity analyses?
3. Would you recommend prioritising this sweep over proceeding directly to thesis finalisation, given the existing five-layer defence?

I am ready to begin the pilot phase immediately upon your confirmation.

Sincerely,

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